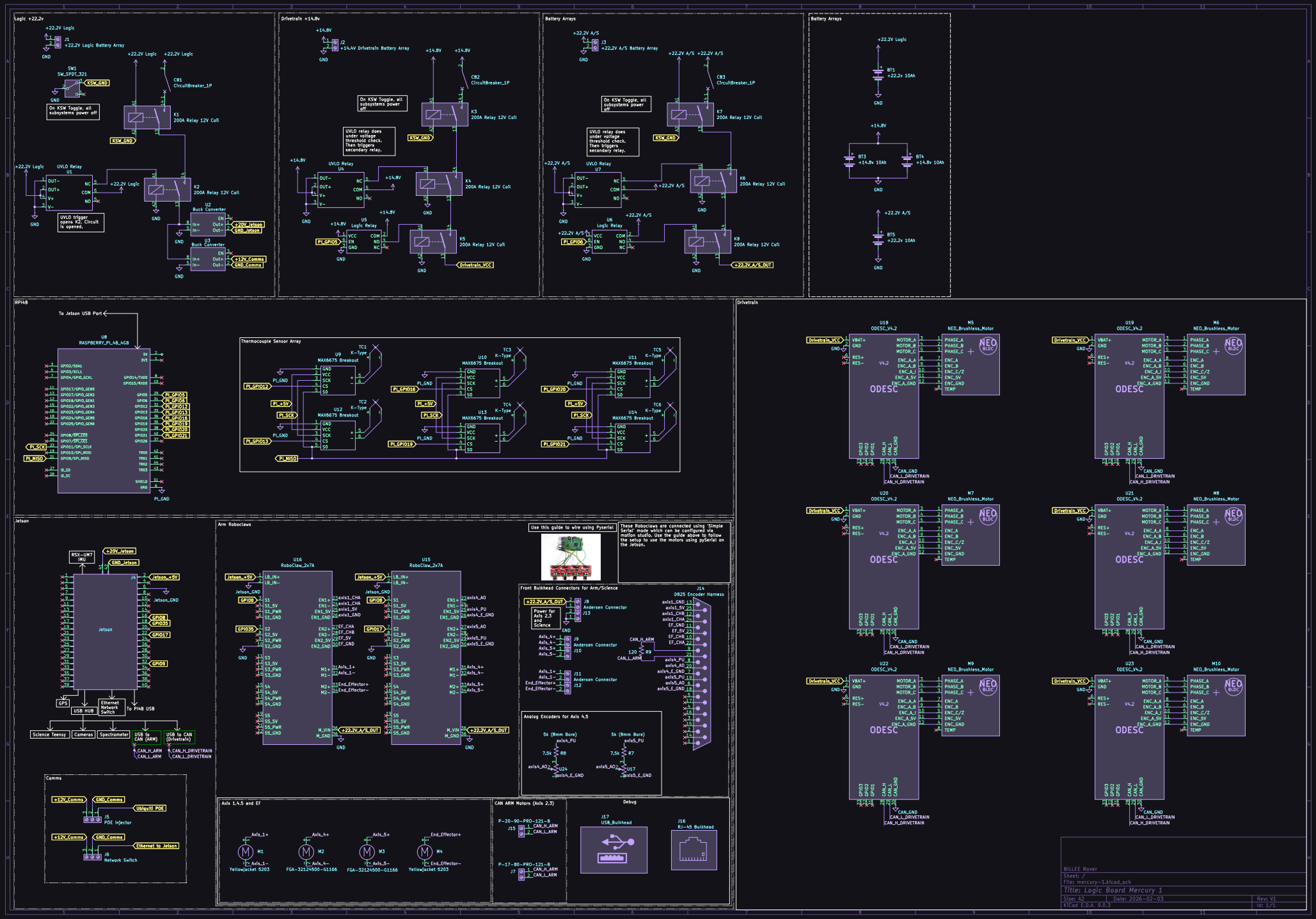
Logic +22.2v
+22.2V Logic
J1
+22.2V Logic Battery Array
GND
SW1
SW_SPDT_321
KSW_GND
On KSW Toggle, all subsystems power off
CB1
CircuitBreaker_1P
K1
200A Relay 12V Coil
UVLO Relay
U1
OUT-
OUT+
V+
V-
NC
COM
NO
UVLO trigger opens K2. Circuit is opened.
K2
200A Relay 12V Coil
U2
Buck Converter
EN
In+
In-
Out+
Out-
+20V_Jetson
GND_Jetson
U3
+12V_Comms
GND_Comms
Drivetrain +14.8v
+14.8V
J2
+14.4V Drivetrain Battery Array
CB2
CircuitBreaker_1P
K3
200A Relay 12V Coil
UVLO relay does under voltage threshold check. Then triggers secondary relay.
U4
K4
200A Relay 12V Coil
U5
Logic Relay
VCC
EN
COM
NO
NC
PL_GPIO5
K5
200A Relay 12V Coil
Drivetrain_VCC
Battery Arrays
+22.2V A/S
J3
+22.2V A/S Battery Array
CB3
CircuitBreaker_1P
K7
200A Relay 12V Coil
U7
K6
200A Relay 12V Coil
U6
PL_GPIO6
K8
200A Relay 12V Coil
+22.2V_A/S_OUT
BT1
+22.2v 10Ah
BT3
+14.8v 10Ah
BT4
+14.8v 10Ah
BT5
+22.2v 10Ah
RPi4B
To Jetson USB Port
U8
RASPBERRY_PI_4B_4GB
PL_SCK
PL_MISO
PL_GND
Thermocouple Sensor Array
U9
MAX6675 Breakout
TC1
K-Type
U12
TC2
U10
TC3
U13
TC4
U11
TC5
U14
TC6
PL_+5V
PL_GPIO12
PL_GPIO13
PL_GPIO16
PL_GPIO19
PL_GPIO26
PL_GPIO21
Drivetrain
U18
ODESC_V4.2
M5
NEO_Brushless_Motor
U19
M6
U20
M7
U21
M8
U22
M9
U23
M10
ODESC
NEO BLDC
Jetson
RSV-UW7 IMU
+20V_Jetson
GND_Jetson
J4
Jetson_+5V
Jetson_GND
GPIO8
GPIO35
GPIO17
GPIO9
GPS
USB HUB
Ethernet Network Switch
To Pi4B USB
Science Teensy
Cameras
Spectrometer
USB to CAN (ARM)
USB to CAN (Drivetrain)
CAN_H_ARM
CAN_L_ARM
CAN_H_DRIVETRAIN
CAN_L_DRIVETRAIN
Comms
+12V_Comms
GND_Comms
J5
PDE Injector
Ubiquiti POE
J6
Network Switch
Ethernet to Jetson
Arm Roboclaws
U16
RoboClaw_2x7A
U15
Use this guide to wire using Pyserial
These Roboclaws are connected using "Simple Serial" mode which can be configured via motion studio. Use the guide above to follow the setup to use the motors using pySerial on the Jetson.
Front Bulkhead Connectors for Arm/Science
+22.2V_A/S_OUT
Power for Axis 2,3 and Science
J8
Andersen Connector
J13
J9
J10
J11
J12
DB25 Encoder Harness
J14
CAN_H_ARM
CAN_L_ARM
R9
Analog Encoders for Axis 4,5
5k (8mm Bore)
R8
R7
U24
U17
Axis 1,4,5 and Ef
M1
M2
M3
M4
Yellowjacket 5203
FGA-32124500-G1166
CAN ARM Motors (Axis 2,3)
P-20-90-PRO-121-B
J15
P-17-80-PRO-121-B
J7
Debug
J17
USB_Bulkhead
J16
RJ-45 Bulkhead
BILLEE Rover
Sheet: /
File: mercury-1.kicad_sch
Title: Logic Board Mercury 1
Size: A2
Date: 2026-02-03
Rev: V1
KiCad E.D.A. 9.0.3
Id: 1/1